

The Ratio-of-Uniforms Idea

Theorem (Kinderman-Monahan, 1977)

Let $h(x) \geq 0$ be proportional to the target density $f(x)$ (i.e. $f(x) = h(x) / \int h$). Define the planar region

$$C = \left\{ (u, v) : x = \frac{v}{u}, 0 \leq u \leq \sqrt{h(x)}, b^- \leq v \leq b^+ \right\}.$$

If (U, V) is drawn **uniformly** over C , then $X = V/U$ has density f .

$$f(x) = \frac{h(x)}{\int h(t) dt}$$

norm. const

$$\int f(x) dx = 1$$

$$\Rightarrow f(x) = \frac{1}{\sqrt{\pi}} e^{-x^2/2}$$

$$\Rightarrow \frac{h(x)}{\int h(t) dt} = \frac{e^{-x^2/2}}{\int_{-\infty}^{\infty} e^{-t^2/2} dt} = \frac{e^{-x^2/2}}{\sqrt{\pi}}$$

$$C = \left\{ (u, v) : x = \frac{v}{u}, \quad 0 < u \leq \sqrt{h(n)}, \quad \frac{1}{5} \leq v \leq \frac{1}{4} \right\}$$

$$u, v \sim U[C] \Rightarrow x = \frac{v}{u} \Rightarrow x \sim f$$

$$f_{u,v}(u,v) = \begin{cases} \frac{1}{|C|} & v, u \\ 0 & \text{o.w.} \end{cases}$$

$$\begin{aligned} u &= \frac{v}{x} \\ v &= xu \\ (u, v) &\xrightarrow{x = \frac{v}{u}} (x, y) \quad \text{where } y = u \end{aligned}$$

$$|J| = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} -\frac{v}{x^2} & 1 \\ u & 0 \end{vmatrix} = u$$

$$f_{x,y}(x,y) = f_{u,v}(u(x,y), v(x,y)) |J|$$

$$f_{x,y}(x,y) = \begin{cases} \frac{1}{|C|} y & 0 < y \leq h(n) \\ 0 & \text{o.w.} \end{cases}$$

✓ $X = \frac{y}{a} \rightarrow$ find density of y
 $0 < y < \sqrt{h(x)}$
 $-\infty < x < \infty$

$$f_{xy}(x, y) = \begin{cases} \frac{y}{|c|} & \sqrt{h(x)} \\ 0 & \text{o.w.} \end{cases}$$

$$f_x(x) = \int_0^{\sqrt{h(x)}} f_{xy}(x, y) dy = \int_0^{\sqrt{h(x)}} \frac{y}{|c|} dy$$

$$f_x(x) = \frac{h(x)}{2|c|} \quad -\infty < x < \infty$$

find c :

$$\int_{-\infty}^{\infty} \int_0^{\sqrt{h(x)}} f_{xy}(x, y) dx dy = 1$$

$$\int_{-\infty}^{\infty} \left(\int_0^{\sqrt{h(x)}} y dy \right) dx = |c|$$

$$\Rightarrow |c| = \int_{-\infty}^{\infty} \frac{h(x)}{2} dx$$

$$f_Y(x) = \frac{h(x)}{2|c|}$$

$$|c| = \frac{\int_{-\infty}^{\infty} h(x) dx}{2}$$

$$f_Y(x) = \frac{h(x)}{\int_{-\infty}^{\infty} h(t) dt} \quad \checkmark$$

↪

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pdf f $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \checkmark C = \left\{ (u, v) : x = \frac{v}{u}, \quad \begin{matrix} 0 \leq u \leq \sqrt{n(x)} \\ \bar{b} \leq v < b^+ \end{matrix} \right\}$$

$$U \sim U[\underbrace{0, \sqrt{n(x)}}_{\text{blue arrow}}], \quad V \sim U[\underbrace{\bar{b}^-, b^+}_{\text{blue arrow}}]$$

Then $X = \frac{V}{U} \sim \underline{f}$

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